National Institute of Technology, Kurukshetra

MID SEMESTER – I, 2025-26

CSIC221 – Machine Learning & Data Analytics

Date: 24/09/2025 (WEDNESDAY)

Closed book, 15 Marks, 50 min

Name: [redacted]

ID: [redacted]

Instruction: Calculators are allowed. Mobile phones and any other electronic gadgets are not allowed.

Q1: A company recorded the quarterly sales (in ₹ lakhs) of its product over 6 quarters. The data is given below:

X (Quarter):	1	2	3	4	5	6
Y (Sales, ₹ lakhs):	48	52	54	58	63	65

Find the following: - (4 Marks)

a) Fit a straight-line trend of the form Y=a+bX using the least squares method.
b) Find the trend equation.
c) Calculate the trend values for each quarter.
d) Estimate the sales for the 7th quarter using your trend equation.

Q2. A small movie platform has the following user–movie rating matrix (rows = users, columns = movies). Ratings are from 1–5, and 0 means not rated. (6 Marks)

$$R = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Answer the following: -

a) Find the singular values of R.
b) Determine the left singular vectors (U) and right singular vectors (V).
c) Write the complete Singular Value Decomposition (SVD) of R.
d) Determine the approximate R_1 matrix using only ONE largest singular value. (Hint: $R_1 = \sigma_1 u_1 v_1^T$)
e) Determine the approximate R_2 matrix using all the largest singular values.
f) Compare the structure of R , R_1, R_2. Comment on the user and movie rating with respect to all R , R_1, R_2 and suggest, which is a better approximation?

Q3. A streaming platform measures a normalized session engagement score X for each user session. The engagement score ranges from 0 (no engagement) to 1 (maximum engagement). Historical analysis suggests the following model for the probability density function of X: (5 Marks)

$$f(x) = x + 1.5\,x^2, \quad 0 \le x \le 1,$$
$$f(x) = 0, \quad \text{otherwise}$$

This model arises because short sessions are common (term x) but there is an increasing tendency towards longer/more engaged sessions (term $1.5\,x^2$).

Answer the following: -

a) Verify that $f(x)$ is a valid PDF on [0,1].
b) Compute the expected value E[X] and the second moment E[X^2].
c) Compute the variance of X, Var(X).